

Assignment 01

Python Programming(CAN303)

Q1.

Write a Python program to accept a student's name, roll number, marks in five subjects, and calculate and display the total marks, percentage, and average. Demonstrate appropriate Python data types.

Q2.

Write a Python program to accept marks of a student and display the grade according to the following criteria:

- 90–100: A+
- 80–89: A
- 70–79: B
- 60–69: C
- 50–59: D
- Below 50: F

Q3.

Write a Python program to accept an integer and determine whether it is a prime number, palindrome, or neither. Use appropriate loops and conditional statements.

Q4.

Create separate functions to calculate the factorial, Fibonacci series, and sum of digits of a given number. Demonstrate function definition, parameters, return values, and function calls.

Q5.

Write a Python program that accepts a string and performs the following operations:

- Count vowels and consonants
- Reverse the string
- Check whether it is a palindrome
- Count the occurrence of a given character

Q6.

Given a list of student marks, write a program to find the maximum, minimum, average, and second-highest marks. Also demonstrate indexing, slicing, insertion, deletion, and sorting operations.

Q7.

Create a student record using a dictionary containing name, roll number, course, and marks. Use a tuple to store subjects and a set to store unique course names. Demonstrate important operations on all three data structures.

Q8.

Given a list of numbers, use lambda functions with `map()` and `filter()` to:

- Generate squares of all numbers
- Extract even numbers
- Extract numbers greater than 50